

Multivariate Bayesian clustering using covariate-informed components with application to boreal vegetation sensitivity

Henry R. Scharf¹  | Ann M. Raiho² | Sierra Pugh³ | Carl A. Roland⁴ |
 David K. Swanson⁵ | Sarah E. Stehn⁴ | Mevin B. Hooten^{2,3,6}

¹ Department of Mathematics and Statistics, San Diego State University, San Diego, California

² Department of Fish, Wildlife, and Conservation Biology, Colorado State University, Fort Collins, Colorado

³ Department of Statistics, Colorado State University, Fort Collins, Colorado

⁴ Denali National Park and Preserve, National Park Service, Fairbanks, Alaska

⁵ Arctic Network, National Park Service, Fairbanks, Alaska

⁶ U.S. Geological Survey, Colorado Cooperative Fish and Wildlife Research Unit, Fort Collins, Colorado

Correspondence

Henry R. Scharf, Department of Mathematics and Statistics, San Diego State University, San Diego, CA.
 Email: hscharf@sdsu.edu

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Abstract

Climate change is impacting both the distribution and abundance of vegetation, especially in far northern latitudes. The effects of climate change are different for every plant assemblage and vary heterogeneously in both space and time. Small changes in climate could result in large vegetation responses in sensitive assemblages but weak responses in robust assemblages. But, patterns and mechanisms of sensitivity and robustness are not yet well understood, largely due to a lack of long-term measurements of climate and vegetation. Fortunately, observations are sometimes available across a broad spatial extent. We develop a novel statistical model for a multivariate response based on unknown cluster-specific effects and covariances, where cluster labels correspond to sensitivity and robustness. Our approach utilizes a prototype model for cluster membership that offers flexibility while enforcing smoothness in cluster probabilities across sites with similar characteristics. We demonstrate our approach with an application to vegetation abundance in Alaska, USA, in which we leverage the broad spatial extent of the study area as a proxy for unrecorded historical observations. In the context of the application, our approach yields interpretable site-level cluster labels associated with assemblage-level sensitivity and robustness without requiring strong *a priori* assumptions about the drivers of climate sensitivity.

KEYWORDS

Alaska national parks, climate change, clustering regression coefficients, community ecology, joint distribution models

1 | INTRODUCTION

Climate change is altering the distribution and abundance of vegetation, and increasing plant vulnerability (Kelly and Goulden, 2008; Allen *et al.*, 2015). Understanding the rate of plant distribution change is integral to predicting future plant distributions and preserving biodiversity (Dawson *et al.*, 2011). However, the response to climate change varies among plant ecosystems (Loarie *et al.*, 2009). Plant sensitivity to climate change conventionally refers to the

response of a plant assemblage relative to the change in climate (Gonzalez *et al.*, 2010). A plant assemblage with high sensitivity to climate will respond more strongly to small changes in the climate, positively or negatively; whereas, a plant assemblage with low sensitivity will respond weakly or not at all to small changes in climate. Different members within a sensitive plant assemblage may respond to changes in climate in different ways; some may increase in abundance and others may decrease in abundance. Determining which plant assemblages may be sensitive versus

robust to climate change is important for assessing ecosystem vulnerability of plant ecosystems to climate changes (Williams *et al.*, 2008; Aubin *et al.*, 2016). In what follows, we generalize the concept of “sensitivity” such that it allows us to identify outliers with respect to robust relationships between response and predictor variables.

Time series data are often used to assess vegetation sensitivity to climate because it is straightforward to calculate the change in vegetation abundance relative to the change in climate over time (Kleinhesselink and Adler, 2018). For example, at global scales, satellites have indicated the sensitivity of vegetation greenness indices to changes in temperature and precipitation worldwide for over 30 years (Seddon *et al.*, 2016). However, global ecosystem modeling studies have shown that vegetation sensitivity to climate is not well understood because of short-term global measurements of temperature and precipitation (Sitch *et al.*, 2008). Furthermore, models of vegetation assemblage dynamics demonstrate that lagged relationships with climate may have large and variable effects on vegetation sensitivity to climate (Peltier *et al.*, 2018), and therefore longer time series are needed to estimate the climate response lags in a given plant assemblage (Blonder *et al.*, 2017). Long time series of plant assemblage composition are typically not available for large spatial domains (although see Kelly *et al.*, 2005), especially in remote regions of arctic and subarctic Alaska. Comprehensive plant assemblage data sets are becoming more available at large spatial scales but not over long time periods (Kao *et al.*, 2012). Thus, a key challenge is to draw information about plant assemblage sensitivity from short-term, large-scale plant assemblage data sets when a large-scale change in vegetation is yet to be perceived in the temporal extent of data.

We present a statistical approach for estimating vegetation sensitivity to climate change from short-term, but large-scale, plant assemblage data sets composed of occupancy and/or abundance measurements for multiple species. We develop a hierarchical mixture model that is able to cluster sites into “sensitive” or “robust” groups based on a generalized concept of sensitivity. The implications of sensitivity and robustness in our model are species specific, however, we model sensitivity at the site level. Therefore, to be appropriate for our joint modeling approach, assemblages should be comprised of species that demonstrate their specialized sensitive responses at the same types of sites (i.e., if one species is exhibiting its signs of sensitivity, they all should be).

We demonstrate the effectiveness of our approach using a large-scale vegetation assemblage data set collected by the National Park Service in Denali National Park and Preserve. Climate change is affecting high latitude regions more than other parts of the world because of Arctic temperature amplification, solar radiation feedback with

decreasing sea ice leading to decreasing solar reflectance. Alaska specifically has seen a 1°C average annual increase in temperature since 1920 with most of the increase in warming occurring during the winter and spring months (Bieniek *et al.*, 2014). This temperature increase has caused vegetation in arctic, subarctic, and boreal ecosystems to be subject to both longer growing seasons (Jia *et al.*, 2003) and widespread increases in forest fires (Turetsky *et al.*, 2011). It has been hypothesized that this high vegetation sensitivity to climate will lead to a catastrophic shift for the boreal forest; changing from a dense conifer forest to a sparse deciduous forest (Scheffer *et al.*, 2012). However, previous analysis of a subset of the plant survey data collected by the National Park Service does not show specific signs of this critical transition (Roland *et al.*, 2019). The statistical framework we developed allows us to better understand the overall vegetation sensitivity to climate in interior Alaska through the analysis of multispecies tree abundance and occupancy data in Denali National Park and Preserve.

2 | METHODS

2.1 | Conditional linear model for response

Throughout the article, we utilize the following common notational conventions. Bold capital letters represent matrices, bold lower case letters represent vectors, and nonbold letters represent scalars, with capital letters reserved for integer-valued variables. We retain the shared letters, possibly with different cases, for subarrays (e.g., β_{lj} is the element in the l th row, j th column of $\mathbf{B}$). Let $\mathbf{Y}$ be an $N \times J$ matrix with elements y_{ij} representing N observations of a continuously valued J -dimensional multivariate response. We specify a linear model for the i th observation, represented by the row vector $\mathbf{y}_i$, conditioned on unobserved cluster labels $z_i \in \{1, \dots, K\}$ as

$$\mathbf{y}_i' | z_i \sim N\left((\boldsymbol{\beta}_0^{(u)}(z_i)' + \mathbf{x}_i^{(u)} \mathbf{B}^{(u)}(z_i) + \mathbf{x}_i^{(s)} \mathbf{B}^{(s)})', \boldsymbol{\Sigma}(z_i)\right). \quad (1)$$

We discuss extensions to non-Gaussian responses in Section 2.4.

In this model, a subset of the regression coefficients and the covariance vary with cluster label. Denote by $\mathbf{X}$ an $N \times L$ matrix corresponding to L covariate values for each observation, and let $\mathbf{X} = (\mathbf{X}^{(s)}, \mathbf{X}^{(u)})$ be a column-wise partition of the matrix into (s)hared and (u)nshared components, with respective numbers of columns given by $L = L^{(s)} + L^{(u)}$. As with $\mathbf{y}_i$, let $\mathbf{x}_i$ represent the i th row of $\mathbf{X}$

corresponding to the covariate values for the i th observation, and define $\mathbf{x}_i^{(s)}$ and $\mathbf{x}_i^{(u)}$ analogously such that $\mathbf{x}_i = (\mathbf{x}_i^{(s)}, \mathbf{x}_i^{(u)})$.

The matrix of covariate effects, $\mathbf{B}^{(u)}(z_i)$, is of dimension $L^{(u)} \times J$, where $L^{(u)}$ is the number of unshared covariates and J is the length of the response vector. Thus, the linear effect of $x_{il}^{(u)}$ on $y_{ij}|z_i = k$ is given by $\beta_{lj}^{(u)}(k)$, the element in the l th row, j th column of $\mathbf{B}^{(u)}(k)$. $\beta_0^{(u)}(z_i)$ is an intercept vector of length J that varies with cluster. We explicitly include the intercept separately from $\mathbf{B}^{(u)}(z_i)$ to facilitate notation of prior distributions. The shared covariate effects, $\mathbf{B}^{(s)}$, are indexed analogously to $\mathbf{B}^{(u)}(z_i)$, but without an index for cluster label. The covariance matrix, $\Sigma(z_i)$, characterizes dependence among the elements of $\mathbf{y}_i$ while accounting for the linear effects of the measured covariates.

We assume for simplicity that the full covariate space is $\mathbb{R}^L$, which is partitioned into the subspaces $\mathbb{R}^{L^{(s)}} \cup \mathbb{R}^{L^{(u)}}$. Thus, the full covariate space is decomposed into shared factors, those covariates whose linear effect on the response is the same regardless of cluster membership, and unshared factors, those covariates that are allowed to vary in their effect based on cluster membership. We defer a discussion of the implications of particular partitions to Section 2.3.

In the application to vegetation abundance in the Central Alaskan network of National Parks, the response vector for a given site has length, J , equal to the number of species under study, and the cluster labels associated with sites that are either robust ($k = 1$) or sensitive ($k = 2$), corresponding to the hypothesis that some locations inherently respond more radically to changes in unshared covariates than others. By allowing for both shared and unshared covariates, it is possible to incorporate existing knowledge about which covariate effects are likely to characterize sensitive versus robust sites.

In general, the cluster labels are guided by informative specifications of the prior distributions for $\mathbf{B}^{(u)}(z_i = k)$ that are distinct for each k . A conjugate form for the prior distributions associated with our model is the multivariate normal distribution, such that $\text{vec}(\mathbf{B}^{(u)}(k)) \sim N(\boldsymbol{\mu}^{(\beta_u)}(k), \boldsymbol{\Sigma}^{(\beta_u)}(k))$ for $k = 1, \dots, K$ (where $\text{vec}(\cdot)$ denotes column-wise vectorization of a matrix), but other multivariate probability distributions could be specified. The multivariate normal prior applies to a length $L^{(u)}J$ vectorization of the coefficient matrices $\text{vec}(\mathbf{B}^{(u)}(k))$. An analogous conjugate prior distribution may be specified for the intercepts and shared effects, such that $\beta_0^{(u)}(k) \sim N(\mu_0^{(\beta_u)}(k), \Sigma_0^{(\beta_u)}(k))$, and $\text{vec}(\mathbf{B}^{(s)}) \sim N(\boldsymbol{\mu}^{(\beta_s)}, \boldsymbol{\Sigma}^{(\beta_s)})$.

For our motivating application, we centered prior distributions for $\mathbf{B}^{(u)}(k)$ at 0 and specified covariances such that

all effects are independent *a priori* with marginal variances $\sigma^{(\beta_u)^2}(k)$ (thus, $\boldsymbol{\Sigma}^{(\beta_u)}(k) = \sigma^{(\beta_u)^2}(k)\mathbf{I}_{L^{(u)}J}$). In accordance with the understanding that a plant assemblage with high sensitivity to climate will respond more strongly to small changes in the climate than an assemblage with low sensitivity, we specified a large variance value for the sensitive cluster, $\sigma^{(\beta_u)^2}(k = 1)$, where small changes in unshared covariates may have a correspondingly large impact on vegetation abundance, and a small variance value for the robust cluster (i.e., $\sigma^{(\beta_u)^2}(k = 2) < \sigma^{(\beta_u)^2}(k = 1)$). Further details of the prior distributions used in the application are available in Appendix A.1.

The latent cluster labels generally define unordered categories distributed according to a categorical distribution as $z_i \sim \text{Cat}(\mathbf{p}_i = (p_{i1}, \dots, p_{iK})')$. The cluster labels can be integrated out to yield a model for $\mathbf{y}_i$ conditioned on the cluster label probabilities given by

$$\mathbf{y}_i' | \mathbf{p}_i \sim \sum_{k=1}^K p_{ik} N\left(\left(\beta_0^{(u)}(z_i) + \mathbf{x}_i^{(u)} \mathbf{B}^{(u)}(z_i) + \mathbf{x}_i^{(s)} \mathbf{B}^{(s)}\right)', \boldsymbol{\Sigma}(k)\right). \quad (2)$$

Thus, the conditional linear model may also be viewed as a mixture model with weights corresponding to the probabilities of cluster membership.

2.2 | Smoothly varying cluster probabilities

A primary goal in our application is to estimate the probabilities, $\mathbf{p}_i$, so that conclusions can be drawn about which sites show evidence of being sensitive or robust. Furthermore, we seek to predict the probability that a new site is sensitive given the observed data. Site sensitivity is expected to be a smooth process with respect to some covariates in the sense that sites with similar landscape covariates tend to share similar ecological characteristics and should therefore have similar sensitivity or robustness. However, the underlying mechanisms that drive sensitivity are not yet thoroughly understood. Thus, we require a model for cluster probabilities that is flexible, incorporates *a priori* knowledge of smoothness in covariate space, and makes mild assumptions about the process generating sensitivity.

We define the cluster probabilities, p_{ik} , using a prototype method (e.g., Hastie *et al.*, 2009) that simultaneously ensures smoothness and allows for prediction of cluster probabilities at new locations. Let $\mathbf{X}^{(p)}$ be a third

covariate matrix (in addition to $\mathbf{X}^{(u)}$ and $\mathbf{X}^{(s)}$) of size $N \times L^{(p)}$, possibly coinciding with elements of $\mathbf{X}$, and let $\mathbf{x}_{i \cdot}^{(p)}$ denote the i th row as before. Let $\mathbf{m}_k \in \mathbb{R}^{L^{(p)}}$ denote a column vector representing the location of a point in this new covariate space. We refer to $\mathbf{m}_k$ as the location of the k th prototype and the elements of $\mathbf{X}^{(p)}$ as the prototype covariates. The probability that the i th site belongs to the k th cluster is defined based on the relative proximity of prototypes to the observed covariate values $\mathbf{x}_{i \cdot}^{(p)}$ such that

$$\alpha_{ik} = \frac{1}{\sqrt{(\mathbf{x}_{i \cdot}^{(p)} - \mathbf{m}_k)' \mathbf{\Lambda} (\mathbf{x}_{i \cdot}^{(p)} - \mathbf{m}_k)}}, \quad p_{ik} = \frac{\alpha_{ik}}{\sum_k \alpha_{ik}}, \quad (3)$$

where $\mathbf{\Lambda}$ is a diagonal matrix of dimension $L^{(p)}$ whose diagonal elements, $\lambda_l > 0, l = 1, \dots, L^{(p)}$, introduce a scalar adjustment to each component of the familiar Euclidean distance (in covariate space). Large values of λ_l emphasize the impact of the l th variable in defining α_{ik} , while small values diminish the impact. A value of $\lambda_l = 0$ effectively removes the l th variable from the prototype space completely.

The role of prototypes is to represent a central hub around which covariate values from respective clusters coalesce. As the distance between the observed covariate values and the k th prototype decreases, α_{ik} increases and the probability that p_{ik} is close to 1 grows as well. As the distance between a prototype and the observed covariate values increases, the opposite occurs. Thus, when prototype locations are far apart, there is a greater potential for a sharp clustering of sites (i.e., cluster probabilities near 0 and 1). When prototype locations overlap, clusters will be less distinguishable.

When the covariates that define the prototype subspace are spatially smooth (as is common with variables measuring landscape characteristics), or when the site coordinates themselves are included as covariates, the prototype method ensures that the cluster probabilities are spatially smooth as well. This propagation of smoothness occurs because of the use of a distance metric (i.e., the weighted Euclidean distance in 3) in the definition of α_{ik} . The triangle inequality requires that two sites with similar covariate values must be located at a similar distance away from each prototype, and will therefore have similar cluster probabilities. Thus, smoothness in this context extends beyond a spatial connotation to include all covariates included in the prototype subspace.

In accordance with our goal of specifying a model for cluster probabilities that is agnostic about the mechanism driving cluster membership, we specify a vague prior for the prototype locations given by $\mathbf{m}_k \sim N(\boldsymbol{\mu}^{(m)}, \sigma^{(m)^2} \mathbf{I})$. Intuitively, the value of $\sigma^{(m)^2}$ should be chosen such

that positions of $\mathbf{m}_k$ simulated from the prior fall among the observed prototype covariates, $\mathbf{X}^{(p)}$. For centered and scaled covariates, specifying a mean-zero Gaussian distribution with variance $\sigma^{(m)^2} = 1$ is reasonable (see Web Appendix A).

2.3 | Regression effects: Three roles

The covariates, $\mathbf{X}^{(s)}$, $\mathbf{X}^{(u)}$, and $\mathbf{X}^{(p)}$, each play an important and distinct role in the clustered regression model given by (1)–(3). As defined in Section 2.1, the collections of shared and unshared covariates, $\mathbf{X}^{(s)}$ and $\mathbf{X}^{(u)}$ are disjoint. Shared effects are constant across clusters, while unshared effects vary by cluster (i.e., the species-specific effects of shared covariates on log-abundance will be the same regardless of a site's latent sensitivity status). Thus, unshared variables are potentially informative about cluster probabilities while shared variables are not.

The extent to which effects differ across clusters provides part of the interpretation behind each cluster. For instance, $\beta_{lj}^{(u)}(k=1) \neq \beta_{lj}^{(u)}(k=2)$ implies that cluster $k=1$ is distinct from $k=2$ in the way the j th response variable changes with respect to the l th unshared covariate. If $\beta_{lj}^{(u)}(k=1) > \beta_{lj}^{(u)}(k=2)$, then the j th variable in sites belonging to the first cluster responds more positively (less negatively) to changes in the l th covariate than does the same variable in sites from the second cluster. An example from our application is the differentiated response to summer total precipitation by tamarack. The effect of increased precipitation (l th covariate) at sensitive sites ($k=2$) was strongly negative for this species ($j=1$), while the effect at robust sites ($k=1$) was weak to negligible (see Figure 2). Thus, unshared covariates assist in defining what it means for a site to belong to one cluster versus another. In the application to vegetation abundance, we hypothesized that the magnitudes of effects for the sensitive cluster would generally be larger than effects in the robust cluster.

Prototype covariates, $\mathbf{X}^{(p)}$, represent a third collection of variables that may include elements from both the shared and unshared sets as well as additional variables. Prototype covariates are potentially informative about cluster probabilities in a different way than unshared covariates. Rather than helping to distinguish the meanings of different clusters, these variables provide information about how likely a site is to belong to each cluster, given the effects $\boldsymbol{\beta}_0^{(u)}$, $\mathbf{B}^{(u)}$, and $\mathbf{B}^{(s)}$. For example, including spatial coordinates as prototype covariates induces spatial smoothness in the cluster probabilities.

Intuitively, covariates belonging to the unshared collection might be expected to impact both the characteristics of cluster labels and the cluster probabilities

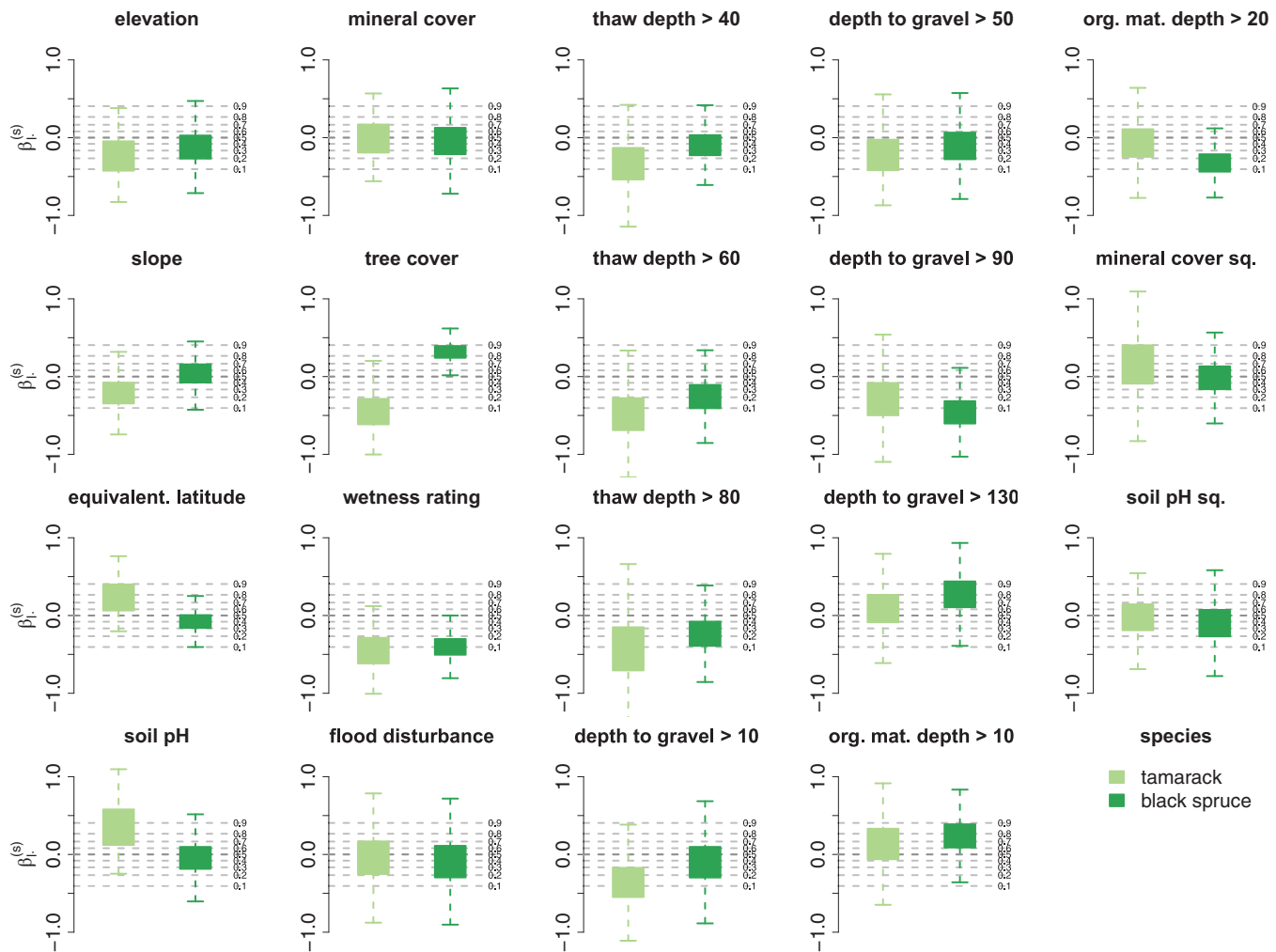


FIGURE 1 Boxplot summaries of posterior distributions for species-specific covariate effects shared across clusters. Gray dashed lines represent prior distribution quantiles 0.1, 0.2, ..., 0.9. This figure appears in color in the electronic version of this article, and any mention of color refers to that version

themselves. Thus, it is natural, although not required, to include all unshared covariates as prototype covariates. In addition, there may be some covariates whose impacts are not expected to vary with cluster label, or may even be independent of the response variable, but are nevertheless informative about cluster probabilities (e.g., the spatial coordinates of a site).

2.4 | Extension to generalized response types

There are two important features of the data used in the application to vegetation in the Alaskan network of National Parks that require special accommodations to implement the model specified in (1)–(3). First, field observations of abundances are potentially made with error. Second, additional binary data were also collected in the field.

Detection/nondetection (1/0) observations for each species were recorded based on censuses of entire plots. To account for uncertainty in the site-level abundance estimates and include the binary detection/nondetection observations, we define $\mathbf{Y}$ as the true log-abundance of each species of vegetation in each plot and extend the model presented in (1)–(3) by adding another layer to the hierarchy that formally couples the latent log-abundance process and the data.

Let $\mathbf{V}$ be an $N \times J$ matrix denoting observations of the true log-abundance, $\mathbf{Y}$, made with uncertainty according to $v_{ij}|y_{ij} \sim N(y_{ij}, \tau_{ij}^2)$. Thus, the observations of log-abundance are assumed to be unbiased with variance that could potentially vary by site and species.

Let $\mathbf{W}$ denote an $N \times J$ matrix of binary random variables in which w_{ij} represents the variable corresponding to the detection/nondetection (1/0) of species j at site i . We model the detection process as $w_{ij} = 1_{\{y_{ij} > d_{ij}\}}$, where d_{ij} is

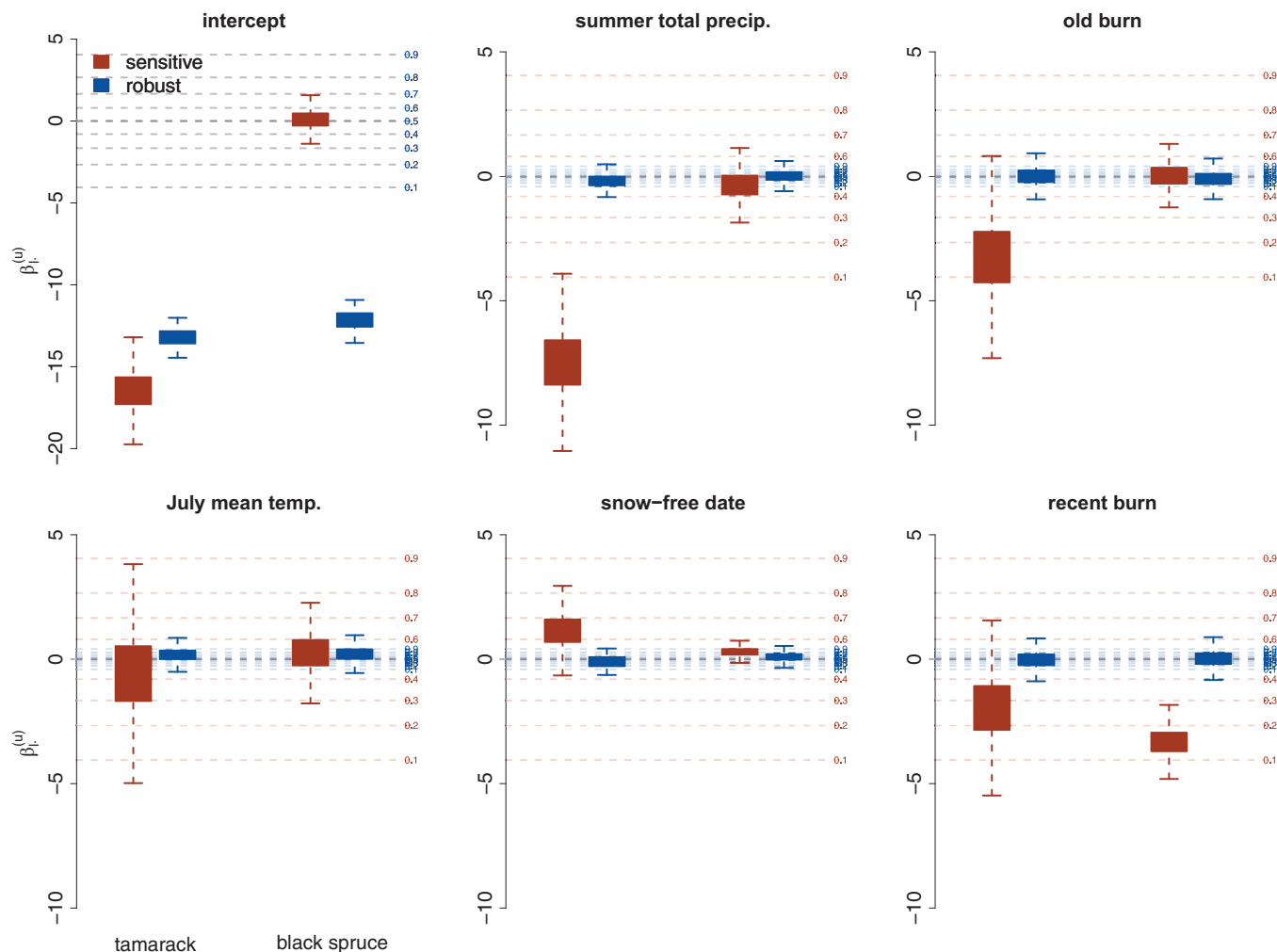


FIGURE 2 Boxplot summaries of posterior distributions for species-specific covariate effects that vary depending on whether a site is sensitive or robust. Dashed lines represent prior distribution quantiles 0.1, 0.2, ..., 0.9 for each cluster. This figure appears in color in the electronic version of this article, and any mention of color refers to that version

a lower detection limit allowed to vary by site and species. The detection limits are assumed to be constant across sites in our application, but could be allowed to vary across sites, potentially according to landscape covariates (e.g., Dorazio and Rodríguez, 2012; Johnson *et al.*, 2013).

The introduction of the detection/nondetection layer in the hierarchical model is particularly valuable for cases when abundance measurements are not available but detection measurements are. Detection information truncates the support of the posterior predictive distribution for log-abundance at the detection threshold and, in general, narrows the width of associated credible intervals. The detection measurements also inform posterior distributions for the effects of shared and unshared covariates on abundance. Detection/nondetection measurements are typically less expensive to collect and hence collected over much larger spatial extents. Thus, the addition of this layer in the model structure permits the inclusion of a large,

potentially informative data source. To facilitate a holistic understanding of the statistical model, Web Appendix A, Figure 1, provides a representation of the directed acyclic graph corresponding to the full hierarchical model with generalized response.

2.5 | Related statistical methods

2.5.1 | Clustering approaches

A broad class of closely related clustering approaches for smoothly varying cluster labels are so-called product partition models (PPMs; Hartigan, 1990). The fundamental assumption of PPMs is that the probability of a particular cluster or partition, ρ , on the indexing set for the data, S_0 , is proportional to the product of K “cohesion” functions operating on the elements in each disjoint cluster,

$S_k \subset S_0$, such that $\Pr(\rho) \propto \prod_{k=1}^K c(S_k)$ and $\cup_{k=1}^K S_k = S_0$. Cohesion functions can be specified that depend on the covariate values for elements in each cluster (Park and Dunson, 2010; Müller *et al.*, 2011), leading to greater probabilities for partitions that vary smoothly in covariate space. While our proposed model for the cluster probabilities is not an example of a PPM, it shares many similar characteristics. Like PPMs, our proposed cluster probability model is an example of a nonparametric model for cluster membership probabilities that induces smoothness through carefully chosen functions of the covariate values.

A related approach for predicting spatially referenced soil organic carbon levels was proposed by Risser *et al.* (2019), who incorporated a regression model conditioned on latent clusters similar to (1). Important distinctions between Risser *et al.* (2019) and the present work are (i) Risser *et al.* (2019) modeled a univariate response; (ii) our approach explicitly accounts for shared and unshared covariate effects, while Risser *et al.* (2019) implicitly treated all covariates as unshared; and (iii) our approach utilizes a process model for the cluster probabilities within a single Bayesian hierarchical model, whereas Risser *et al.* (2019) employed model averaging to account for uncertainty in the cluster probabilities. Another related approach for areal spatial processes was proposed by Kazor and Hering (2019), who adopted the same linear structure conditioned on a spatially smooth clustering process. Neither Kazor and Hering (2019) nor Risser *et al.* (2019) accommodated generalized response types, although, in principle, our proposed extensions could be used analogously for either case.

Kleinhesselink and Adler (2018) also studied differentiated responses of vegetation to changing climate based on sensitivity groups using an auto-regressive model with carefully chosen interaction terms. Their analysis revealed how vegetation in sites near the edges of habitable zones (as defined through covariate values) may have specialized responses to changes in climate. Our approach is fundamentally different in how sites are identified as sensitive. We infer sensitivity from the data through the specification of informative priors on covariate effects, and thus, sensitive sites need not be associated with the edges of habitable zones.

2.5.2 | Joint species distribution models (JSDMs)

Our proposed model is an example of a JSDM, for which a rich body of research has been developed (e.g., Pollock *et al.*, 2014). The unifying motivation for JSDMs is explaining the variation in abundance and/or occupancy patterns of several species, while accounting for dependence

in species' responses. Frequently, subgroups of an assemblage can be usefully clustered based on their responses patterns (e.g., Hui *et al.*, 2015; Johnson and Sinclair, 2017; Taylor-Rodriguez *et al.*, 2017). For example, Hui *et al.* (2015) explored two penalization schemes for fitting finite mixture regression models to presence-absence patterns of 200 species in the Great Barrier Reef. Johnson and Sinclair (2017) used an explicit clustering model that grouped species into an unknown number of guilds consistent across sites.

A fundamental difference between our approach and other models for joint species distribution modeling is that we cluster sites, rather than species. Our motivating data set has far fewer species than most applications in the JSDM literature, so accounting for species-specific covariate effects does not incur the usual implementation challenges. In addition, our scientific interests are different from those typically motivating JSDM research. We are interested in identifying locations where an assemblage demonstrates evidence of stronger or weaker responses to climate, whereas most JSDMs are used to identify subsets of an assemblage that response similarly to exogenous covariates. Although sensitivity and robustness may manifest themselves in different ways for different species, we hypothesize that, where one species is responding in its own sensitive or robust way to climate, other species in our assemblage are responding in theirs.

3 | APPLICATION: WOODY VEGETATION ABUNDANCE

3.1 | Data

The specific data we analyzed were collected by the Central Alaska Network in Denali National Park and Preserve. We focus on two tree species of crucial interest in Alaskan National Parks: black spruce (*Picea mariana*) and tamarack (*Larix laricina*). In general, sensitivity and robustness of vegetation are best understood as an assemblage-based, rather than strictly site-based characteristic; different collections of species may correspond to different maps of sensitivity. We identified black spruce and tamarack as co-occurring species expected to exhibit similar sensitivity patterns in space.

It has been suggested previously that some areas of Alaska formerly dominated by coniferous vegetation, such as black spruce and tamarack, are susceptible to replacement by deciduous vegetation after fire or insect disturbance (Chapin and Starfield, 1997), which can have global climatic impacts (Bonan *et al.*, 1992). While these changes have been observed in some areas of Alaska, it is not apparent that the replacement of dense spruce forests is

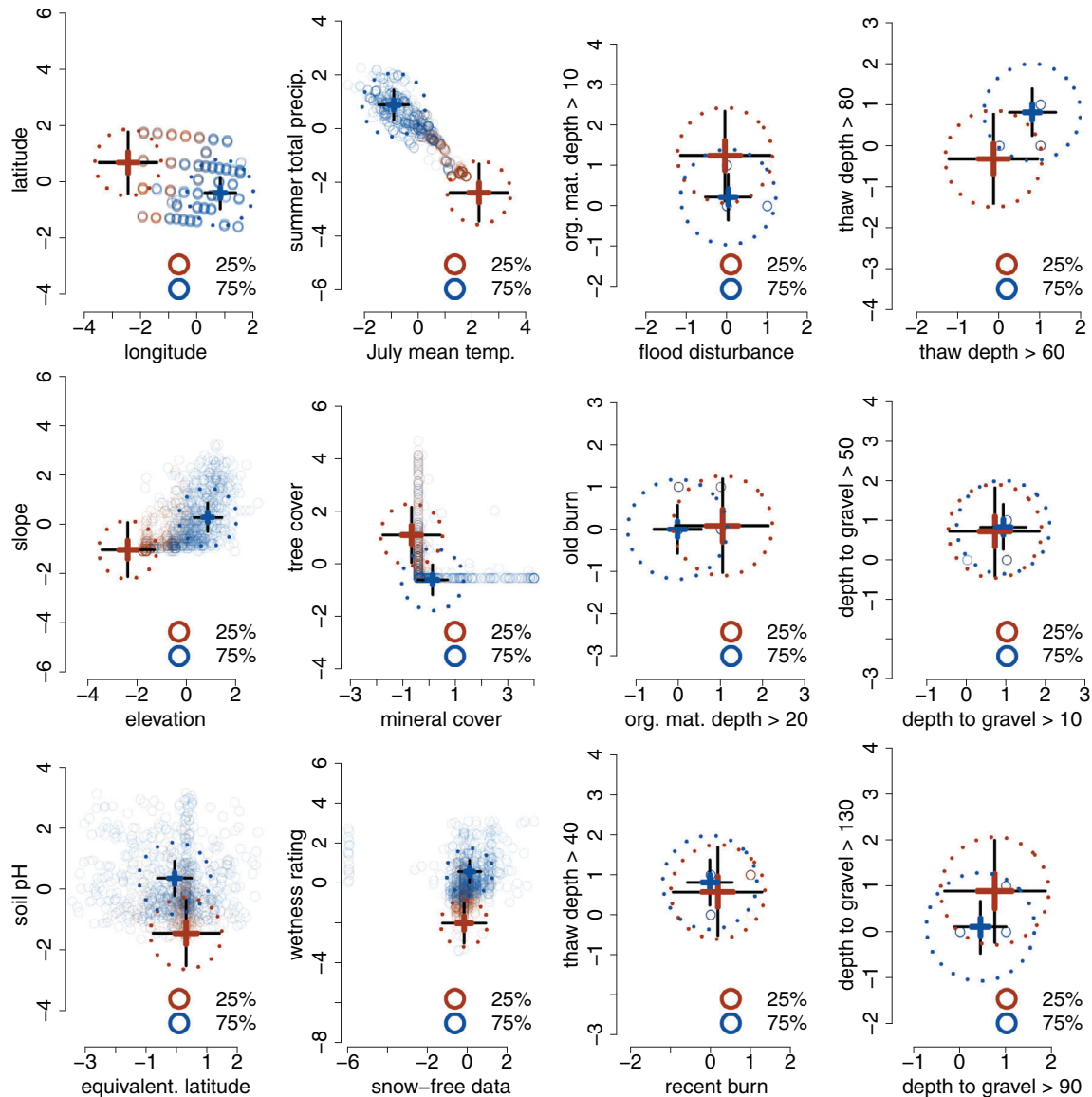


FIGURE 3 Prototype location distributions are summarized marginally by pairs. Red/blue line segments give 50% equal-tailed credible intervals for sensitive/robust sites, and black line segments give 95% equal-tailed credible intervals. Red/blue dashed lines show circles with radius equal to $\lambda_i = 1$. Observed covariate values are plotted in the background of each figure colored according to $\bar{z}_i(1) < 0.5$ and $\bar{z}_i(1) > 0.5$ for sensitive and robust sites, respectively. This figure appears in color in the electronic version of this article, and any mention of color refers to that version

occurring throughout Alaskan National Parks (Roland *et al.*, 2019).

Data were collected using an unstratified, multistage systematic grid design. The present application utilizes measurements of species occurrence and basal area (BA; i.e., abundance) at a total of 999 sites in Denali National Park and Preserve (Figure 4). Plant species occurrence was censused at all sites. BA for each species was measured at all sites and included stems that reached breast height (1.37 m, DBH). A total of 79 BA measurements for tamarack and 215 measurements for black spruce were made. There were 74 sites with recorded BA measurements for

both species, and thus 779 sites with occurrence measurements but no BA measurements. Table 1 provides the complete frequency information with respect to occupancy and the existence of BA measurements, and Web Appendix A, Figure 2 provides a visual representation. Site-referenced observations are based on the sum of all individual tree BA measurements.

Several physical covariates were also measured at each site, including: slope angle and azimuth, elevation, topographic position, disturbance regime, slope shape, drainage characteristics, and evidence of frost action and fire. Soil attributes at each site were also recorded using

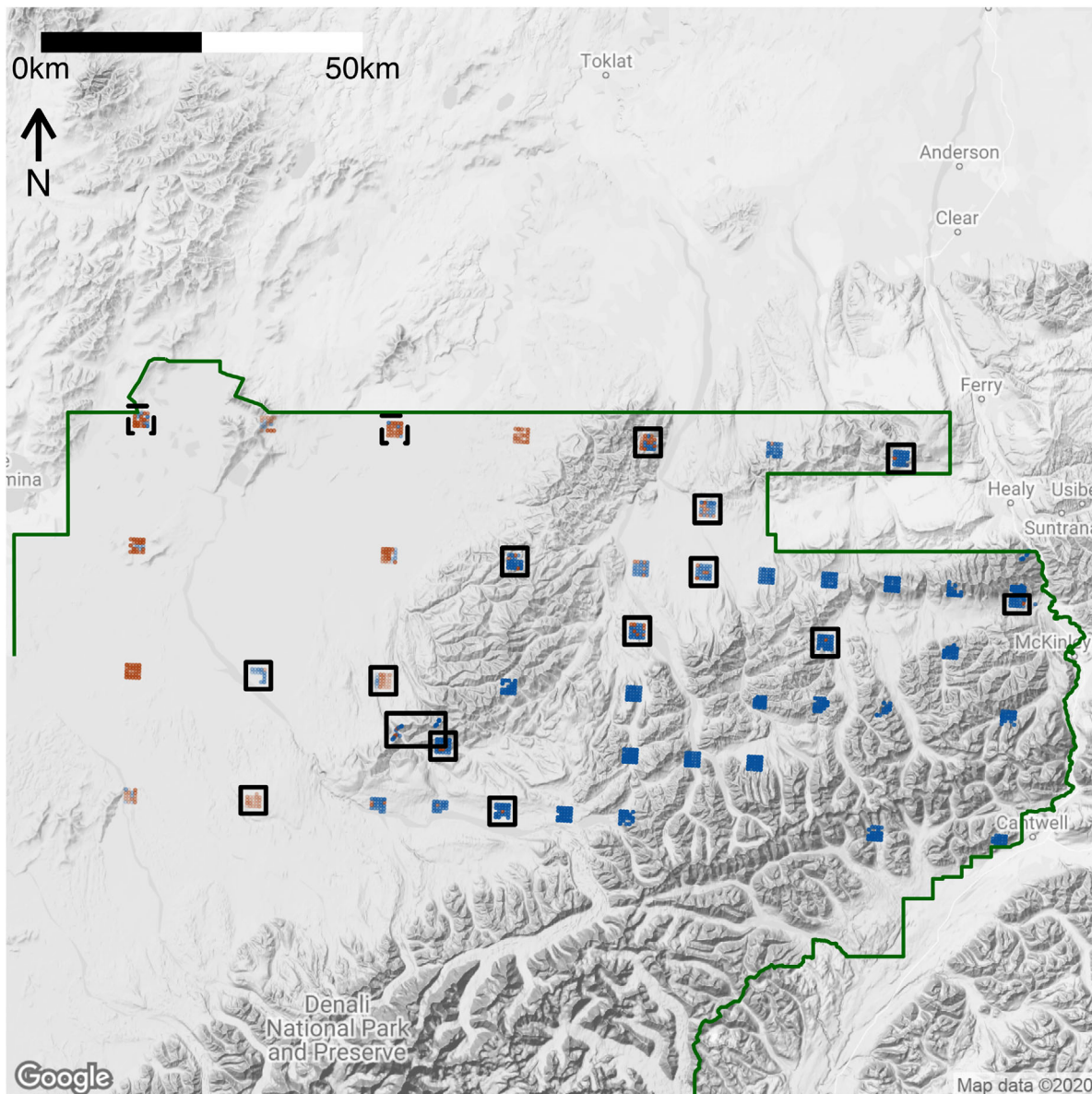


FIGURE 4 Sensitive (red) and robust (blue) sites as determined by $\bar{z}_i(1) < 0.5$ and $\bar{z}_i(1) > 0.5$, respectively. Transparency depicts uncertainty of cluster label, with values of $\bar{z}_i(1)$ near 0 or 1 represented as more opaque and values near 0.5 more transparent. Solid boxes indicate grids containing at least one site where one or more species were present, yet predicted to be absent. Dashed boxes indicate grids containing at least one plot where one or more species were absent, yet predicted to be present. The green line shows the north eastern boundary of Denali National Park and Preserve. This figure appears in color in the electronic version of this article, and any mention of color refers to that version

soil cores in four quadrants of each site. A full account of the rationale and implementation of this sampling design can be found in Roland *et al.* (2004).

3.2 | Implementation

We fit the model to observations of occurrence and log-BA of black spruce and tamarack in Denali National Park and Preserve. As mentioned, the first cluster ($z_i = 1$) corresponded to a sensitive cluster, while the second cluster

($z_i = 2$) corresponded to a robust cluster. The distinction was aided by the specification of prior variance–covariance matrices for the regression coefficients as

$$\begin{aligned}\Sigma_0^{(\beta_u)}(1) &= \Sigma_0^{(\beta_u)}(2) = 10\mathbf{I}_J, & \Sigma^{(\beta_u)}(1) &= 10\mathbf{I}_{L(u)}, \\ \Sigma^{(\beta_u)}(2) &= 10^{-1}\mathbf{I}_{L(u)}.\end{aligned}\quad (4)$$

Marginal variances for the unshared intercepts, $\Sigma_0^{(\beta_u)}(1)$ and $\Sigma_0^{(\beta_u)}(2)$, were equal for both clusters because we had no *a priori* reason to believe overall abundance should

TABLE 1 Frequencies for all combinations of occupancy and record of BA measurement by species. Entries with an upright font style give the frequencies associated with the corresponding row and column. Entries with stylized fonts correspond to *unions*, $\cup$, and *intersections*, $\cap$, of both species for the indicated subscript on the set operator. For example, $\cup_{Occ}$ indicates the aggregated frequencies satisfying the requirement that *at least* one species matches the specified occupancy status, subject to other relevant row and column information, whereas $\cap_{Occ}$ indicates that **both** species match the specified occupancy status. The value of 74 in the first numerical row and third numerical column therefore indicates that there were 74 sites in the data set for which both species were observed to be present and a BA measurement was made for each species. Each entry in the table corresponds to a combination of 0, 2, or 3 unions and/or intersections

	Present				Absent				$\cup_{Occ}$	
	Tam.	Bl. sp.	$\cap_{Occ}$	$\cup_{BA}$	Tam.	Bl. sp.	$\cap_{Occ}$	$\cup_{BA}$	$\cap_{BA}$	$\cup_{BA}$
BA	79	215	74	101	0	0	0	0	74	220
no BA	29	40	6	101	891	744	743	743	779	925
$\cup_{BA}$	108	255	80	107	891	744	743	743	853	999

TABLE 2 Covariates with an asterisk* were binned and treated as categorical. Column headings correspond to shared (s), unshared (u), and prototype (p). See Web Appendix A for details

Covariate	(s)	(u)	(p)
intercept		✓	
burn status: <i>unburned</i> (> 80 years), <i>old</i> (20–80 years), <i>recent</i> (<20 yrs)		✓	✓
depth to gravel*	✓		✓
elevation	✓		✓
equivalent latitude	✓		✓
flood disturbance	✓		✓
july mean temp.		✓	✓
latitude			✓
longitude			✓
mineral cover	✓		✓
(mineral cover) ²	✓		
organic mat. depth*	✓		✓
slope	✓		✓
snow-free date		✓	✓
soil pH	✓		✓
(soil pH) ²	✓		
summer total precip.		✓	✓
thaw depth*	✓		✓
tree cover	✓		✓
wetness rating	✓		✓

differ by cluster. Prior distributions for all other parameters were specified to be weakly informative (see Web Appendix A for a full description of all priors). We grouped covariates into shared, unshared, and prototype categories (Table 2) based on expert knowledge of the ecosystem.

We used the package “nimble” (de Valpine *et al.*, 2017) for the R statistical computing environment to construct Markov chain Monte Carlo (MCMC) samplers to fit the cluster-based regression model to data. The MCMC algorithm we implemented was based on the Rao-Blackwellized version of the cluster-based regression

model in 2. Code to reproduce our analyses is available in Web Appendix B.

3.3 | Results

3.3.1 | Covariate effects

Figure 1 shows box plots for posterior samples of $\mathbf{B}^{(s)}$, the effects shared across cluster labels. For example, in the subfigure for soil pH (row 4, column 1), the distinct linear

effects for tamarack (left) and black spruce (right) are visible, with tamarack showing a preference for more basic soil. Other strong effects for at least one species include tree cover, slope, wetness, organic material depth, and thaw depth.

Figure 2 shows box plots of the posterior distributions for the unshared (i.e., cluster-specific) effects. For example, in the subfigure for summer total precipitation (row 1, column 2), we can see a marked difference in the response of certain vegetation to increasing summer precipitation at sensitive sites compared to robust sites. In particular, increasing summer precipitation is associated with a decrease in abundance for tamarack at sensitive sites compared to robust ones. Similarly, a recent burn was not associated with a large change in abundance of black spruce at robust sites, while it was associated with a large decrease in abundance at sensitive sites (row 2, column 3).

3.3.2 | Sensitive versus robust clusters

We obtained samples from the posterior distribution for z_i in the original formulation of the model for log-abundance given in (1) through composition sampling. The distinction between $\mathbf{p}_i$ and the expected value of z_i is subtle. The mixture probabilities, $\mathbf{p}_i$, describe the probability that a site with landscape characteristics, $\mathbf{x}_i$, will be either sensitive or robust. The observed data inform $\mathbf{p}_i$ through the regression coefficients and prototype locations. The result is a global description of cluster label trends for particular combinations of characteristics. In contrast, the expected value of z_i is the local probability that site i belongs to a given cluster after further taking into account how the measurements, $\mathbf{y}_i$, align with the distributions of log-abundance implied by each cluster. Examining posterior distributions of both $\mathbf{p}_i$ and z_i can provide useful information.

We denote by $\bar{z}_i(k)$ the proportion of realizations from the posterior distribution for which $z_i = k$. Thus, $\bar{z}_i(1)$ is the posterior probability that site i is a sensitive site, and $\bar{z}_i(2) = 1 - \bar{z}_i(1)$ is the posterior probability that it is robust. For simplicity, we refer to sites as sensitive/robust when $\bar{z}_i(1) > /< 0.5$.

Table 3 shows the number of sensitive/robust sites for sites occupied by 2, 1, or 0 of the study species, and the mean value of p_{i1} within each combination. The vast majority (98%) of sites occupied by at least one species indicated sensitivity (i.e., $\bar{z}_i(1) > 0.5$), and all unoccupied sites indicated robustness. Thus, sensitive/robust clusters tended to divide along occupied/unoccupied sites. There were four robust sites occupied by a single species, which in all cases was black spruce. Sites that are both robust and occupied are of particular scientific interest because they

TABLE 3 Summary of inferred sensitivity, $\bar{z}_i(1) > 0.5$, and occupancy w_{ij} partitioned by the number of species occupying each site (2, 1, or 0). The integer value is the number of robust/sensitive sites, and the value in parentheses is the mean value of p_{i1} corresponding the combination of occupation and sensitivity

No. species	Robust	Sensitive	Total
2	0 (NA)	107 (0.67)	107 (0.67)
1	4 (0.37)	105 (0.47)	149 (0.47)
0	743 (0.20)	0 (NA)	743 (0.20)
Total	747 (0.20)	252 (0.56)	999 (0.29)

represent locations where vegetation is both available to study, and predicted to remain stable in abundance in the near future.

Figure 3 shows pairwise plots that summarize the posterior distributions of the prototypes and the way prototype covariates inform cluster probabilities. Pairs were chosen based on covariates suspected of interaction. For example, from the subfigure in row 1, column 2, we can see that sites with warmer, drier summers are more likely to be sensitive. The prototype covariates that most strongly influenced the probability of belonging to a sensitive/robust cluster were longitude, elevation, summer total precipitation, soil pH, July mean temperature, summer total precipitation, tree cover, and wetness rating, which is visible in Figure 3 as larger distances between prototypes for these variables.

Figure 4 shows each site location labeled according to the inferred highest probability cluster: red for sensitive, blue for robust. The transparency of each point corresponds to the degree of certainty in the mode assignment, with darker colors indicating posterior cluster probabilities, $\bar{z}_i(1)$, heavily weighted toward a single cluster and lighter colors corresponding to more balanced cluster probabilities. This map reinforces information contained in Figure 3. For example, several high elevation, steeply sloped sites in Denali National Park and Preserve show a high probability of being robust.

3.3.3 | Changes in occupancy

For some combinations of site and species, model-based predictions of occupancy status diverged from observations. To summarize differences between predicted and observed occupancy, we categorized site/species combinations with posterior predictive probabilities of occupancy below 0.3 as “predicted to be unoccupied,” and site/species combinations with posterior predictive probabilities of occupancy above 0.7 as “predicted to be occupied.” Boxes in Figure 4 indicate grids of sites where the predicted occupancy of at least one site/species pair differed from the observed status.

Observed and predicted occupancy status can diverge for several reasons including imperfect model fit. Another plausible explanation is that a site's occupancy status is out of alignment because it is on the verge of a change in occupancy. Occupied sites predicted to be unoccupied may have new, local conditions that are no longer suitable for the proliferation of certain species and are currently occupied by legacy populations. Analogously, unoccupied sites predicted to be occupied may be ripe for colonization.

There were 30 species/site combinations (out of 1998) predicted to be unoccupied but observed to be occupied, corresponding to 29 unique, sensitive locations and 0 robust locations. A total of 10 combinations corresponded to tamarack, leaving 20 corresponding to black spruce. Recorded observations of log-abundance were made at 5 and 15 sites for tamarack and black spruce, respectively. In general, medians of the posterior predictive distributions for true log-abundances were much lower than observed log-abundances (Web Appendix A, Figure 5). This discrepancy is consistent with the hypothesis that one or both species may abruptly become absent from these locations after the departure of large, legacy individuals. There were four sites predicted to be occupied by black spruce where that species was absent (no such sites for tamarack), and all four sites were estimated to be robust. Medians of the posterior predictive distributions of true log-abundance for these sites were well above the detection threshold (Web Appendix A, Figure 5), consistent with the hypothesis that black spruce would be highly successful at these sites and might therefore be likely to colonize them.

4 | DISCUSSION

The statistical model we developed and applied to observations of vegetation abundance and presence/absence offers several opportunities for scientific learning compared to commonly used linear models. Traditional spatial modeling approaches account for missing covariates that could explain variation in the response through the inclusion of separable spatial random effects, which only provide a phenomenological explanation for residual positive dependence in the response variables. In contrast, our approach is consistent with a more mechanistic, scientifically informed hypothesis that certain sites may be more likely to respond more strongly to unshared covariates like spatially heterogeneous climate change, and offers the potential for identifying and even predicting the locations of sensitive sites. In addition, our approach integrates two different data types and naturally accommodates missing observations.

Our proposed model incorporates pathways for complex, nonlinear mechanisms that link limiting factors like

soil acidity to vegetation abundance. Although, in our analysis, the effect of soil acidity did not vary with cluster (i.e., soil pH was a shared covariate), because soil pH was also included as a prototype covariate, the probability of a site being labeled sensitive or robust depended on soil pH and sites with higher pH were more likely to be robust (Figure 3, row 3, column 1). Black spruce had a lower intercept associated with robust sites compared to sensitive ones (Figure 2, row 1, column 1). Thus, increasing soil pH was indirectly associated with a decrease in the expected abundance of black spruce through the clustering process.

Our application demonstrates how our model can provide scientific learning about changing vegetation assemblages in arctic and subarctic regions. We considered only a single collection of species defined *a priori*, however, a useful extension to our approach would include a framework for identifying important groups of species from within a larger assemblage that can be expected to share sensitive/robust labels. A model selection perspective might offer one successful approach for identifying "optimal" subsets of species. Alternatively, it could be useful in some applications to compare cluster labels for two overlapping collections of species, in which case the meaning of cluster labels would be interpreted conditional on the species in the subgroup.

Our modeling approach can also be used in a variety of other applications. Specifically, there are similarities between our ecological application and some case-control studies of human health. For example, in some situations, health status is measured for a representative subset of individuals from a study population, but longitudinal data on the individuals are not available. Medical researchers may wish to account for genetic effects (shared) while inferring sensitivity to environmental pollution (unshared). In such cases, the use of a snapshot of data from a population that exhibits heterogeneity in genetics could provide insights about possible effects of pollution on various groups within the larger population.

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DATA AVAILABILITY STATEMENT

The data that support the findings in this paper are available in the Supporting Information of this article (Web Appendix B).

ORCID

Henry R. Scharf  <https://orcid.org/0000-0002-9266-4191>

REFERENCES

- Allen, C.D., Breshears, D.D. and McDowell, N.G. (2015) On underestimation of global vulnerability to tree mortality and forest die-off from hotter drought in the anthropocene. *Ecosphere*, 6, 1–55.
- Aubin, I., Munson, A., Cardou, F., Burton, P., Isabel, N., Pedlar, J et al. (2016) Traits to stay, traits to move: a review of functional traits to assess sensitivity and adaptive capacity of temperate and boreal trees to climate change. *Environmental Reviews*, 24, 164–186.
- Bieniek, P.A., Walsh, J.E., Thoman, R.L. and Bhatt, U.S. (2014) Using climate divisions to analyze variations and trends in Alaska temperature and precipitation. *Journal of Climate*, 27, 2800–2818.
- Blonder, B., Moulton, D.E., Blois, J., Enquist, B.J., Graae, B.J., Macias-Fauria, M et al. (2017) Predictability in community dynamics. *Ecology Letters*, 20, 293–306.
- Bonan, G.B., Pollard, D. and Thompson, S.L. (1992) Effects of boreal forest vegetation on global climate. *Nature*, 359, 716.
- Chapin, F.S. and Starfield, A.M. (1997) Time lags and novel ecosystems in response to transient climatic change in arctic Alaska. *Climatic Change*, 35, 449–461.
- Dawson, T.P., Jackson, S.T., House, J.I., Prentice, I.C. and Mace, G.M. (2011) Beyond predictions: biodiversity conservation in a changing climate. *Science*, 332, 53–58.
- de Valpine, P., Turek, D., Paciorek, C., Anderson-Bergman, C., Temple Lang, D. and Bodik, R. (2017) Programming with models: writing statistical algorithms for general model structures with NIMBLE. *Journal of Computational and Graphical Statistics*, 26, 403–417.
- Dorazio, R.M. and Rodríguez, D.T. (2012) A Gibbs sampler for Bayesian analysis of site-occupancy data. *Methods in Ecology and Evolution*, 3, 1093–1098.
- Gonzalez, P., Neilson, R.P., Lenihan, J.M. and Drapek, R.J. (2010) Global patterns in the vulnerability of ecosystems to vegetation shifts due to climate change. *Global Ecology and Biogeography*, 19, 755–768.
- Hartigan, J.A. (1990) Partition models. *Communications in Statistics - Theory and Methods*, 19, 2745–2756.
- Hastie, T., Tibshirani, R., Friedman, J. and Franklin, J. (2009) Prototype methods and nearest-neighbors. In *The Elements of Statistical Learning*, chapter 13, 2nd edition. New York, NY: Springer.
- Hui, F.K., Warton, D.I., Foster, S.D., et al. (2015) Multi-species distribution modeling using penalized mixture of regressions. *The Annals of Applied Statistics*, 9, 866–882.
- Jia, G.J., Epstein, H.E. and Walker, D.A. (2003) Greening of arctic Alaska, 1981–2001. *Geophysical Research Letters*, 30, 1–4.
- Johnson, D.S., Conn, P.B., Hooten, M.B., Ray, J.C. and Pond, B.A. (2013) Spatial occupancy models for large data sets. *Ecology*, 94, 801–808.
- Johnson, D.S. and Sinclair, E.H. (2017) Modeling joint abundance of multiple species using Dirichlet process mixtures. *Environmetrics*, 28, e2440.
- Kao, R.H., Gibson, C.M., Gallery, R.E., Meier, C.L., Barnett, D.T., Docherty, K.M et al. (2012) Neon terrestrial field observations: designing continental-scale, standardized sampling. *Ecosphere*, 3, 1–17.
- Kazor, K. and Hering, A.S. (2019) Mixture of regression models for large spatial datasets. *Technometrics*, 61, 507–523.
- Kelly, A.E. and Goulden, M.L. (2008) Rapid shifts in plant distribution with recent climate change. *Proceedings of the National Academy of Sciences*, 105, 11823–11826.
- Kelly, M., Allen-Diaz, B. and Kobzina, N. (2005) Digitization of a historic dataset: The Wieslander California vegetation type mapping project. *Madroño*, 52, 191–201.
- Kleinhesselink, A.R. and Adler, P.B. (2018) The response of big sagebrush (*Artemisia tridentata*) to interannual climate variation changes across its range. *Ecology*, 99, 1139–1149.
- Loarie, S.R., Duffy, P.B., Hamilton, H., Asner, G.P., Field, C.B. and Ackerly, D.D. (2009) The velocity of climate change. *Nature*, 462, 1052.
- Müller, P., Quintana, F.A. and Rosner, G.L. (2011) A product partition model with regression on covariates. *Journal of Computational and Graphical Statistics*, 20, 260–278.
- Park, J.-H. and Dunson, D.B. (2010) Bayesian generalized product partition model. *Statistica Sinica*, 20, 1203–1226.
- Peltier, D.M., Barber, J.J. and Ogle, K. (2018) Quantifying antecedent climatic drivers of tree growth in the southwestern US. *Journal of Ecology*, 106, 613–624.
- Pollock, L.J., Tingley, R., Morris, W.K., Golding, N., O'Hara, R.B., Parris, K.M et al. (2014) Understanding co-occurrence by modelling species simultaneously with a Joint Species Distribution Model (JSDM). *Methods in Ecology and Evolution*, 5, 397–406.
- Risser, M.D., Calder, C.A., Berrocal, V.J. and Berrett, C. (2019) Non-stationary spatial prediction of soil organic carbon: implications for stock assessment decision making. *Annals of Applied Statistics*, 13, 165–188.
- Roland, C.A., Oakley, K., Debevec, E. and Loomis, T. (2004) Monitoring vegetation structure and composition at multiple scales in the central alaska network. In NPS technical report CAKN-001. National Park Service Fairbanks, Alaska, USA.
- Roland, C.A., Schmidt, J.H., Winder, S.G., Stehn, S.E. and Nicklen, E.F. (2019) Regional variation in interior Alaskan boreal forests is driven by fire disturbance, topography and climate. *Ecological Monographs*, 89, e01369.
- Scheffer, M., Hirota, M., Holmgren, M., Van Nes, E.H. and Chapin, F.S. (2012) Thresholds for boreal biome transitions. *Proceedings of the National Academy of Sciences*, 109, 21384–21389.
- Seddon, A.W., Macias-Fauria, M., Long, P.R., Benz, D. and Willis, K.J. (2016) Sensitivity of global terrestrial ecosystems to climate variability. *Nature*, 531, 229.
- Sitch, S., Huntingford, C., Gedney, N., Levy, P., Lomas, M., Piao, S et al. (2008) Evaluation of the terrestrial carbon cycle, future plant geography and climate-carbon cycle feedbacks using five dynamic global vegetation models (DGVMs). *Global Change Biology*, 14, 2015–2039.
- Taylor-Rodriguez, D., Kaufeld, K., Schliep, E.M., Clark, J.S., Gelfand, A.E., et al. (2017) Joint species distribution modeling: dimension reduction using Dirichlet processes. *Bayesian Analysis*, 12, 939–967.
- Turetsky, M.R., Kane, E.S., Harden, J.W., Ottmar, R.D., Manies, K.L., Hoy, E et al. (2011) Recent acceleration of biomass burning and carbon losses in Alaskan forests and peatlands. *Nature Geoscience*, 4, 27.
- Williams, S.E., Shoo, L.P., Isaac, J.L., Hoffmann, A.A. and Langham, G. (2008) Towards an integrated framework for assessing the vulnerability of species to climate change. *PLoS Biology*, 6, e325.

SUPPORTING INFORMATION

Web Appendix A, referenced in Sections 2.2, 2.4, 3.1, 3.2, and 3.3.3, and Web Appendix B, referenced in Section 3.2, are available with this paper at the Biometrics website on Wiley Online Library. Web Appendix B contains the data that support the findings in this paper.

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